



Hantro VC9000D Video Decoder

Test Bench User Manual

Document Revision 1.02

3 January 2024

This document is compatible with the following product:

Hantro VC9000D software release version 2.4.80

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Preface

This chapter introduces the *Hantro VC9000D Video Decoder Test Bench User Manual*.

About This Document

This document describes the test bench released for Hantro VC9000D multi-format video decoder IP and provides simulation guidelines.

Additional Reading

- Hantro VC9000D Video Decoder Hardware Features
- Hantro VC9000D Video Decoder Memory Buffer and Format Organization
- Hantro VC9000D Video Decoder API User Manual
- Hantro VC9000D Video Decoder Wrapper Layer API User Manual

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1 Overview

This document describes the Application Programming Interface (API) of the Hantro VC9000D hardware based decoder hardware.

Details of the Hantro Video Decoder API are described in this document.

- First, components such as data types, return codes, enumerations and relevant structures are described.
- Then, function syntax and description is presented.
- Finally, a programming overview and examples are presented.

The code is written in C and parameter types follow standard C conventions. This document assumes that the reader understands the fundamentals of the C-language and the VCDEC standards.

Currently, there are no deprecated functions in this API.

1.1 Supported Standards

The API discussed in this version of the document is compatible with the following video coding standards and profiles. Additional levels may be supported via software.

2 Options

The chapter describes the options available in the test bench for configuring the video decoder.

- To specify a long option, separate the option and the value with an equal sign (=). For example, `--num-pictures=10`.
- To specify a short option, separate the option and the value with an equal sign (=) or omit the equal sign. For example, `-N10`.

In this chapter:

- In short options, [s] requires a string value and [n] requires an integer value.
- The value range of an option is provided at the beginning of its description, with the boundaries separated by a couple of periods . Both boundaries are inclusive.
- The default value of an option is marked by parentheses (default) at the end of the first paragraph of the option description.

2.1 Help Info Options

Short	Long Option	Description
-H	--help	Print command line parameters help info.

2.2 Testbench Options

Short	Long Option	Description
	--tb-cfg	Specify the path of tb.cfg file, e.g. ./tb.cfg, e.g. ./xxx.cfg

2.2.1 Input Options

Short	Long Option	Description
-i	--input-format	Force input file format interpretation. Format can be one of the following. bs - bytestream format ivf - IVF file format webm - WebM file format
-p	--packet-by-packet	Packetize input bitstream
-F	--full-stream	Read full-stream into single buffer. Only with bytestream.
-u	--nalu	NALU input bitstream (without start code)
-N	--num-pictures	Stop after outputting n frames.
-n	--non-ringbuffer	Disable ringbuffer mode for stream input buffer. (Default enabled)

2.2.2 Threading Options

Short	Long Option	Description
-Z	--separate-output-thread	Run output handling in separate thread.
	--multimode=[n]	Temporary testing multiple decoder instances in multi-thread mode or multi-process mode(only valid for g2dec). Specify decoders running in parallel in multi-thread or multi- process mode. 0 - Disable(default) 1 - Multi-thread mode 2 - Multi-process mode (reserved)
	--streamcfg [s]	Specify the filename storing decoder options.

2.2.3 Output Options

Short	Long Option	Description
	-Cd	Output crop picture by testbench instead of PP.
-t	--trace-files	Generate hardware trace files.
-r	--rtl-trace	Trace file format for RTL (extra CU ctrl).
-R	--disable-display-order	Output in decoding order.
-X	--no-write	Disable output writing.
-O	--output-file	Write output to the specified file.
-Q	--single-frames-out	Output single frames.
-M	--md5	Write MD5 sum to output instead of yuv.
-m	--md5-per-pic	Write MD5 sum for each picture to output instead of yuv. Note: -m (-M) not guarantee sync up with the third md5 tools.

2.2.4 Logmsg Options

If logging is enabled by setting LOGMSG=y when you compile the test bench, you can configure the log output by

setting the following environment variables in Linux Bash before running the test bench.

- The following example commands enable decoder IRQ logs and register configuration logs of the all information to be output.

E.g. `--logtracemap=IRQ|REGS --logoutlevel=4`

Short	Long Option	Description
	--logoutlevel=[n]	Control log message output level. 0 - quiet 1 - error (default) 2 - info 3 - debug 4 - all

Short	Long Option	Description
	--logtracemap=[s]	Indicate what kind of log should be output for debug. E.g. --logtracemap=API... CFG - config (dump full command line feature list and tb.cfg) API - api (dump api call and some key param) (default) DWL - dwl (dump dwl call, replace -D_DWL_DEBUG) MEM - mem (dump memory usage) DPB - dpb (replace -DDPB_LOCK_TRACE) REGS - regs (dump registers) VCMD - vcmd (dump vcmd instruction) IRQ - irq (dump IRQ status) PERF - perf (performamnce information) ALL - all (dump all trace)
	--logoutdir=[s]	Control log message output dir 0 - all log output to stdout (default) 1 - all log use one log file 2 - each thread of instance has its own file 3 - all log output to stderr

2.3 Decoding Flow Options

Short	Long Option	Description
	--low-latency	Enable low latency platform running flag.
	--secure	Enable secure mode flag.
	--mc	Enable multi-core decoding (frame-based for H264/HEVC/AVS2, tile-based for AV1/VP9).
	--intra-only	Decode intra frames only.
	--partial	Enable partial decoding. Not supported by VC9000D.

2.4 Driver Selection Options

Short	Long Option	Description
	--dec-dev	Set decoder device file name. [/tmp/dev/hantrodec]
	--mem-dev	Set memalloc device file name. [/tmp/dev/memalloc]

2.5 Decoding Options

Short	Long Option	Description
-b	--compress-bypass	Bypass reference frame compression.
	--num-buffers=[n]	Set [n] frame buffers when initialization and then start decoding. Valid values: [1, 15]
	--skip-frame=[s]	Skip some frames when decoding. Valid values: non_ref_recon - Decode all frame, but decoder HW doesn't write recon data of non-reference frames (only hevc & h264_high10) non_ref - Don't decode non-reference frames. none - Close this feature (don't skip anything).
	--index-file=[s] -A[n]	Set the length of data that demuxer transmits to decoder via index file. Set [path/file]s as the path of index file. Use macro USE_DUMP_INPUT_STREAM to get index file. Set stride aligned to n bytes. Valid values: 1/8/16/32/64/128/256/512/1024/2048
	--align-height=[n]	Set height aligned to n bytes. Valid values: 1/8/16/32/64/128/256/512/1024/2048
	--auxinfo	Specify auxiliary info dumping type(HEVC/H264). 0 - Disable(Default) 1 - QP dumping 2 - MV dumping 3 - Both QP and MV dumping
	--tlayer	Maximum Temporal Layer to be decoded. -1 to decode all temporal layers. Valid values: [-1, 7]

2.5.1 Error Concealment Options

There are three typical software error concealment policies. User can choose an appropriate policy base on your situation.

- Focus on the clarity of the video, just display the correct pictures. Use --ec=3 to enable this mode.
- Focus on the fluency of the video, display all pictures including error pictures. Use --ec=2 to enable this mode.

- Both clarity and fluency of the video are considered(default), user can set a tolerable error ratio by parameters.

Short	Long Option	Description
	--error-ratio=[n]	Tolerable error ratio [n], the frame with an error ratio less than or equal to n% will be output. Only used in [-ec=tolerant_error] mode.
	--ec	Set error handling policy: tolerant_error - Balanced strategy ([1], Default) ignore_error - Fluency first strategy [2] no_error - Clarity first strategy [3]
	--dis-ec	Disable HW error concealment (HEVC/H264).
	--dis-slice	Disable slice decoding (HEVC/H264).

2.6 PP Options

Short	Long Option	Description
	--pp [n]	Enable pp [n].
	--dec400_enable -E[s]	Enable dec400 of current pp channel if has dec400 ip,default bypass. Enable hw feature where feature is one of the following: rs - Raster scan conversion pack10 - YUV420pack10 for 10-bit and 12-bit stream p010 - P010 format for 10-bit and 12-bit stream l010 - l010 format for 10-bit and 12-bit stream L010 - L010 format for 10-bit and 12-bit stream 1010 - 1010 format for 10-bit and 12-bit stream
	--crop [wxh@(x,y)]	Enable cropping from (x,y) with size wxh.
	--scale [wxh]	Enable down scaling to 1/x, 1/y in two directions.
	--scale [-dx:y]	Enable down scaling to 1/x, 1/y in two directions.

Short	Long Option	Description
-d	--down_scale -A[n] -Dwxh -C[xywh]NNN	Enable down scale feature. Downscale is one of the following: a). [ds_ratio] [1, 2, 4, 8] Down scale to 1/[ds_ratio] in both directions b). [ds_ratio_x]:[ds_ratio_y] [2, 4, 8] Down scale to 1/[ds_ratio_x] in horizontal, 1/[ds_ratio_y] in vertical PP output size wxh. E.g., -D1280x720 : PP output size 1280x720 Cropping parameters. (--crop [xywh]NNN). E.g., -Cx8 -Cy16 : Crop from (8, 16) -Cw720 -Ch480 : Crop size 720x480
-f	--force-8bits	Force output in 8 bits per pixel for 10-bit stream.
	--cr-first	PP outputs chroma in CrCb order
	--pp-shaper	Enable shaper for pp <n>. (Default support)
	--pp-shaper-dis	Disable shaper for pp <n>.
	--pp-shaper-no-pad	Shaper no need to pad for frame right edge.
	--delogo=[01]	Specify the delogo area. There are two delogo area supported. Need to configure following parameters for a given delogo area:
	--pos=wxh@(x,y)	Set delog rectangle. E.g., --pos=36x24@(64,48)
	--show[n]	Show the delogo border.
	--mode[s]	Select the delogo mode. Valid string: PIXEL_INTERPOLATION - Delogo with interpolated pixels. PIXEL_REPLACE - Delogo with given pixels from option --YUV
	--YUV=(r,g,b)	Set the replacing value if PIXEL_REPLACE mode is enabled. E.g., --YUV=(113,142,129)
	--pp-ycbcr	Set the pp output chroma format. By default CbCr are stored in one plane (a.k.a., semi-planar format) if "--pp-planar" is not enabled explicitly. Valid values: YUV420 YUV422 YUV444

Short	Long Option	Description
-a	--pp-planar	PP output in planar format. Default is semi-planar.
-U	--pp-tiled-out	PP output in tile format. Default is 4x4 tile.
	--pp-luma-only	PP output YUV400.
	--second-crop	Enable the crop after scale, the crop parameter should be followed.
	--pp-comp	Enable TILED16x16 or TILED32x8 compression in PP.
	--tiled-mode[s]	Specify the tile mode/size. Always enabled for FBC. Valid values: TILED8x8 TILED16x16 TILED128x2 TILED64x64(super_tile: store RGB data)
	--pp-packed-mode -s[yc][n]	Specify the YUV packed mode for customized format. Valid values: YUYV UYVY Set PP stride for y/c plane to [n] (out_stride [yc][n]). E.g., -sy720 -sc360 Set ystride and cstride to 720 and 360.
	--lc-stripe[n]	Set the lines number to trigger line counter interrupt. n = 0 to disable line count update.

2.6.1 YUV2RGB Options

Short	Long Option	Description
	--pp-rgb	Enable compact YUV2RGB.
	--pp-rgb-planar	Enable planar YUV2RGB, with each component in separated planes.
	--rgb-fmat	Set the RGB output format. Valid Values: RGB888/BGR888 R16G16B16/B16G16R16 ARGB888/ABGR888/RGBA888/BGRA888 (Planar not supported) A2R10G10B10/A2B10G10R10 (Planar not supported) R10G10B10A2/B10G10R10A2 (Planar not supported) XRGB888/XBGR888 (Planar not supported)

Short	Long Option	Description
	--rgb-std	Set the standard coeff to do YUV2RGB. Valid values: BT601 BT601_L BT709 BT709_L BT2020 BT2020_L
	--rgb-alpha	Set the alpha value.

2.6.2 PP Filter Options

Short	Long Option	Description
	--pp-filter=[s]	Specify filtering algothrim for scaling. Valid Values: VSI_LINEAR LANCZOS NEAREST BILINEAR BICUBIC SPLINE BOX FAST_LINEAR FAST_BICUBIC
	--filter-param=[mxn]	Specify the fitering window size mxn. (ONLY 2x2 is allowed).
	--pp-src-sel=[n]	Specify how to select the source for scaling. Valid values: DOWN_ROUND NO_ROUND UP_ROUND
	--pp-pad-yuv=[Y,U,V]	Set the values for source pixels out of picture when scaling.
	--antialias	Fixed bicubic or bilinear filter size. (valid value: 0, 1)

2.6.3 PPDEC RFC Input Options

Short	Long Option	Description
	--IL	Specify input luma RFC data.
	--IC	Specify input chroma RFC data.
	--SL	Set input luma RFC data stride. Luma_stride calculation: align = alignment of input data pixel_width = (bit_depth_luma == 8 && bit_depth_chroma == 8) ? 8 : 10; luma_stride = NextMultiple(8 * frame_width * pixel_width , align * 8) / 8;
	--SC	Set input chroma RFC data stride. Chroma_stride calculation: align = alignment of input data pixel_width = (bit_depth_luma == 8 && bit_depth_chroma == 8) ? 8 : 10; chroma_stride = NextMultiple(4 * frame_width * pixel_width, align * 8) / 8;
	--LT	Specify input luma table file. Luma table size calculation: pic_width_in_cbs = (pic_width + 8 - 1) / 8; pic_height_in_cbs = pic_height / 8; table_stride = NextMultiple(pic_width_in_cbs, 16); luma_table_size = table_stride * pic_height_in_cbs;
	--CT	Specify input chroma table file. Chroma table size calculation: pic_width_in_cbs = (pic_width + 16 - 1) / 16 pic_height_in_cbs = pic_height / 8 ; table_stride = NextMultiple(pic_width_in_cbs, 16); chroma_table_size = table_stride * pic_height_in_cbs;
	--AT	Specify input luma table + input chroma table.
	--LB	Specify input luma bitdepth.
	--CB	Specify input chroma bitdepth.

2.7 Format Specific Options

This section describes the specific options of specific formats.

2.7.1 H264 Only Features

The video decoder supports multi view sequence, it's enabled by setting `--mvc` in command line.

It's also supported sequence which is flexible macroblock ordering or abitray slice ordering. You should use

`--is-asofmo` to specify this sequence has this characteristic so that the decoder can do related process.

Short	Long Option	Description
	<code>--mvc</code>	Enable MVC decoding (H264 only).
	<code>--is-asofmo</code>	Input file may include ASO/FMO feature.

2.7.2 AV1 Only Features

The options only supported in av1 are listed as follows:

Short	Long Option	Description
	<code>--nottranspose</code>	Don't transpose tile processing order. Default transopse.
	<code>--oppoint</code>	Just decode operating point 0.

2.7.3 JPEG Only Features

The options only supported in jpeg are listed as follows:

Short	Long Option	Description
	<code>--full-only</code>	Force to full resolution decoding only.
	<code>--ri-mc-enable</code>	Enable restart interval based multicore decoding.
	<code>--instant-buffer</code>	Output buffer provided by user. (History Customized) (replace <code>--instant_buffer</code> with <code>--instant-buffer</code>)

2.7.4 VP8 Only Features

The options only supported in vp8 are listed as follows:

Short	Long Option	Description
	--snap-shot	Set decode format to webp.
	--extra-bits	Add n bytes of extra space after stream buffer for decoder.
	--user-mem-alloc	User allocates picture buffers.

2.7.5 VP6 Only Features

The options only supported in vp6 are listed as follows:

Short	Long Option	Description
	--alpha	Stream contains alpha channel. (replace -A with --alpha, only --alpha is available)

2.7.6 MPEG4 Only Features

The options only supported in mpeg4 are listed as follows:

Short	Long Option	Description
	--strm-sorenson	Decode Sorenson Spark stream.
	--strm-custom	Decode DivX4 or DivX5 stream.
	--custom	Decode DivX3 stream of resolution widthxheight.

2.7.7 VC1 Only Features

The options only supported in vc1 are listed as follows:

Short	Long Option	Description
	--long-stream	Enable support for long streams.
	--frame-picture	Enable frame picture writing in multiresolutin output.

2.7.8 RV Only Features

The options only supported in rv are listed as follows:

Short	Long Option	Description
	--rv-input	Input file in RealVideo format.

2.8 PPDEC Only Features

The options only supported in pp standalone are listed as follows:

2.8.1 Input Options

Short	Long Option	Description
	-S[n]	Set input stream stride to n bytes.
	-W[n]	Set input stream width to n pixels (MUST).
	-H[n]	Set input stream height to n pixels (MUST).
	--inp010	Specify input YUV in p010 format (10-bit pixel in MSB of 16 bits).
	--inl010	Specify input YUV in l010 format (10-bit pixel in LSB of 16 bits).
	--pix-fmt	Specify input YUV pixel format (NV12, 420P, TILED4x4).

2.8.2 RGB2YUV Options

Short	Long Option	Description
	--in-rgb-fmt	Set the RGB input format. (conflict with option "--rgb-fmt") Only support packed mode. Valid Values: RGB888/BGR888 ARGB888/ABGR888 A2R10G10B10/A2B10G10R10 X2R10G10B10/X2B10G10R10

Document Revision History

This section describes top level differences in the versions of this document.

Note

This document is not necessarily updated for each patch or minor revision. The information in this document tends to be stable across a revision (nnn) series.

Document Revision	Date	Compatible Core	Description
1.02	2024-01-02	VC9000D	Updated the document name and template.
1.01	2023-12-21	VC9000D	Initial release.